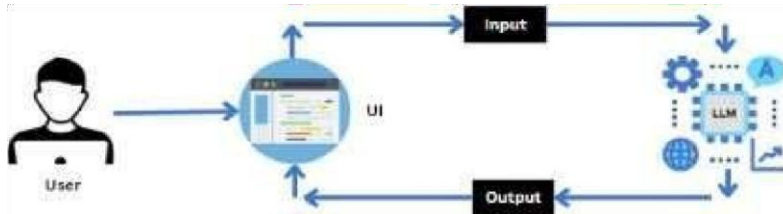


Project Design Phase

Solution Architecture

Date	27 February 2026
Team ID	LTVIP2026TMIDS84546
Project Name	Explore with AI: Custom Itineraries for Your Next Journey
Maximum Marks	2 Marks

Solution Architecture:



1. Overview

Explore with AI is a web and mobile-based platform that allows users to enter travel preferences (destination, duration, interests) and receive a personalized, AI-generated travel itinerary. The system is designed for simplicity, speed, and scalability.

2. High-Level Components

A. User Interface (Frontend)

- Technology: Streamlit (Python) or React.js for web/mobile interface
- Functionality:
 - Input fields for destination, trip duration, and preferences/interests
 - Submit button to generate itinerary
 - Display of AI-generated itinerary in a structured, day-wise format
 - Options to review, customize, or export itinerary (PDF/Text)
 - Display user-friendly error messages

B. API Layer / Backend

- Technology: Python (Flask/FastAPI) Functionality:
 - Receives user input from frontend
 - Validates inputs (destination, dates, preferences) ◦ Handles requests to the AI model ◦ Returns structured itinerary results to frontend

C. AI Engine

- Technology: Gemini Pro LLM / OpenAI GPT / other generative AI model
 - Functionality:
 - Processes input parameters (destination, duration, interests) ◦ Generates detailed day-wise itinerary including:
 - Attractions and ☐ sightseeing
 - Local dining ☐ recommendations
 - Suggested activities and ☐ timing
 - ☐ Travel tips and optional hidden gems ◦ Ensures structured output for easy rendering on frontend

D. Data Layer (Optional)

- Technology: Cloud database (MongoDB/PostgreSQL) ☐
 - Functionality: ◦ Stores predefined destination data (optional)
 - Stores user-generated itineraries (if history/export feature is implemented) ◦ Logs analytics for system monitoring

E. Export & Reporting Module

- Functionality:
 - Convert itinerary to PDF or downloadable formats ◦ Copy/export to clipboard
 - Optional integration with email or messaging for sharing

3. Key Features of Architecture

- Modular Design: Frontend, backend, AI engine, and data layer are loosely coupled.
- Scalable: Can handle more destinations, multiple users, and AI requests simultaneously.
- Extensible: Future features can include multi-destination planning, travel booking integrations, or content generation for blogs.
- Simple Workflow: Focused on input → AI processing → output, minimizing user friction.